$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos\left(k \frac{x\pi}{L}\right) + \sum_{k=1}^{\infty} b_k \sin\left(k \frac{x\pi}{L}\right)$$

$$a_k = \frac{1}{L}\int_{-L}^{L} f(x) \cos\left(k \frac{x\pi}{L}\right) dx$$

$$b_k = \frac{1}{L}\int_{-L}^{L} f(x) \sin\left(k \frac{x\pi}{L}\right) dx$$

F. cosine series

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos\left(k \frac{x\pi}{L}\right)$$

$$a_k = \frac{1}{L}\int_{-L}^{L} f(x) \cos\left(k \frac{x\pi}{L}\right) dx$$

$$= \frac{2}{L}\int_{0}^{L} f(x) \cos\left(k \frac{x\pi}{L}\right) dx$$

Fourier Sine Series

$$f(x) = \sum_{k=1}^{\infty} b_k \sin\left(k \frac{x\pi}{L}\right)$$

$$b_k = \frac{1}{L}\int_{-L}^{L} f(x) \sin\left(k \frac{x\pi}{L}\right) dx$$

$$b_k = \frac{2}{L}\int_{0}^{L} f(x) \sin\left(k \frac{x\pi}{L}\right) dx$$

$$\mathcal{F}\{f(x)\} = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$\mathcal{F}^{-1}\{f(\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\omega) e^{i\omega x} d\omega$$

Fourier Cosine Transform

$$\mathcal{F}_c\{f(x)\} = \int_{0}^{\infty} f(x) \cos(\omega x) dx$$

$$\mathcal{F}_c^{-1}\{f(\omega)\} = \frac{2}{\pi}\int_{0}^{\infty} f(\omega) \cos(\omega x) d\omega$$

Fourier Sine Transform

$$\mathcal{F}_s\{f(x)\} = \int_{0}^{\infty} f(x) \sin(\omega x) dx$$

$$\mathcal{F}_s^{-1}\{f(\omega)\} = \frac{2}{\pi}\int_{0}^{\infty} f(\omega) \sin(\omega x) d\omega$$

Example

$$f(x) = \begin{cases} 1, & 0 \le x < 1 \\ 0, & x \geq 1 \end{cases}$$

Show that,

$$\int_0^\infty \frac{\sin^3 x}{x}\, dx = ?$$

Answer:

$$F_s\{f(x)\} = \int_0^\infty f(x) \sin(wx)\, dx$$

$$= \underbrace{\int_0^1 (1) \sin(wx)\, dx}_{} + \int_1^\infty 0 \sin(wx)\, dx$$

$$= -\frac{\cos(wx)}{w}\Big|_0^1 = -\frac{\cos(w)}{w} - \left(-\frac{1}{w}\right)$$

$$= \frac{1}{w}\left(1 - \cos(w)\right) = \frac{1}{w}\left(1 - \cos\left(2 \times \frac{w}{2}\right)\right)$$

$$= \boxed{\frac{1}{w}\, 2\sin^2\left(\frac{w}{2}\right)}$$

Step-2

$$f(x) = F_s^{-1}\left\{\frac{2\sin^2\left(\frac{w}{2}\right)}{w}\right\}$$

$$= \frac{2}{\pi}\int_0^\infty \frac{2\sin^2\left(\frac{w}{2}\right)}{w} \sin(wx)\, dw$$

$$= \frac{4}{\pi}\int_0^\infty \frac{\sin^2\left(\frac{w}{2}\right) \sin(wx)}{w}\, dw$$

Step-3

Let, $x = \frac{1}{2}$,

$$\underbrace{f\left(\frac{1}{2}\right)} = \frac{4}{\pi}\int_0^\infty \frac{\sin^2\left(\frac{w}{2}\right)\sin\left(\frac{w}{2}\right)}{w}\,dw$$

$$\Rightarrow 1\times\frac{\pi}{4} = \boxed{\int_0^\infty \frac{\sin^3\left(\frac{w}{2}\right)}{w}\,dw}$$

$$\frac{w}{2} = u \Rightarrow w = 2u$$

$$\frac{dw}{du} = 2 \Rightarrow dw = 2du$$

$$\Rightarrow \frac{\pi}{4} = \int_0^\infty \frac{\sin^3 u}{\cancel{2}u}\,\cancel{2}du$$

$$\Rightarrow \frac{\pi}{4} = \int_0^\infty \frac{\sin^3 u}{u}\,du \quad \text{let } u = x, \quad \boxed{\int_0^\infty \frac{\sin^3 x}{x}\,dx = \frac{\pi}{4}}$$

Example

$f(x) = e^{-x}$, $x \geq 0$, show that, $\boxed{\int_0^\infty \frac{\cos(mx)}{1+x^2}\,dx = \frac{\pi}{2}e^{-m}}$

Step-1

$$F_c\{f(x)\} = \int_0^\infty f(x)\cos(wx)\,dx$$

$$= \underbrace{\int_0^\infty \underbrace{e^{-x}}_{v}\,\underbrace{\cos(wx)}_{u}\,dx}_{I}$$

	u	v
+	$\cos(wx)$	e^{-x}
−	$-w\sin(wx)$	$-e^{-x}$
+	$-w^2\cos(wx)$	e^{-x}

$$= \underbrace{\left[-\cos(wx)e^{-x} + w\sin(wx)e^{-x}\right]_0^\infty}_{\text{normal}} - \underbrace{w^2\int_0^\infty \cos(wx)e^{-x}\,dx}_{\text{Extra (repeated math)}}$$

$$= 0 - (-1.1 + 0) - w^2 I$$

$$I = 1 - w^2 I \Rightarrow I + w^2 I = 1$$

$$\Rightarrow I(1+w^2) = 1$$

$$\boxed{I = \frac{1}{1+w^2}}$$

Step-2

$$f(x) = \mathcal{F}_c^{-1}\left\{\frac{1}{1+w^2}\right\} = \frac{2}{\pi}\int_0^\infty \frac{1}{1+w^2}\cos(wx)\,dw$$

let, $x = m$,

$$f(m) = \frac{2}{\pi}\int_0^\infty \frac{\cos(mw)}{1+w^2}\,dw$$

$$e^{-m} \times \frac{\pi}{2} = \boxed{\int_0^\infty \frac{\cos(mw)}{1+w^2}\,dw}$$

$w = x$,

$$\boxed{\frac{\pi}{2}e^{-m} = \int_0^\infty \frac{\cos(mx)}{1+x^2}\,dx}$$

$$\int_0^1 x\,dx = \frac{x^2}{2}\Big|_0^1 = \frac{1}{2}$$

$$\int_0^1 w\,dw = \frac{w^2}{2}\Big|_0^1 = \frac{1}{2}$$

dummy variable

Example

$f(x) = e^{-mx}$, $x \geq 0$, Show that,

$$\int_0^\infty \frac{\beta\cos(\rho v)}{v^2+\beta^2}\,dv = \frac{\pi}{2}e^{-\rho\beta}, \quad \beta, \rho > 0$$

Step-1

$$\mathcal{F}_c\{f(x)\} = \int_0^\infty f(x)\cos(wx)\,dx$$

$$= \underbrace{\int_0^\infty \underbrace{e^{-mx}}_{v} \underbrace{\cos(wx)}_{u}\, dx}_{I}$$

sign	u	v
+	$\cos(wx)$	e^{-mx}
−	$-w\sin(wx)$	$\frac{e^{-mx}}{-m}$
+	$-w^2\cos(wx)$	$\frac{e^{-mx}}{m^2}$

$$= \left[\cos(wx)\frac{e^{-mx}}{-m} + w\sin(wx)\frac{e^{-mx}}{m^2}\right]_0^\infty - \int_0^\infty \frac{w^2}{m^2}\cos(wx)e^{-mx}\,dx$$

$$= 0 - \left(-\frac{1}{m} + 0\right) - \frac{w^2}{m^2}\underbrace{\int_0^\infty \cos(wx)e^{-xm}\,dx}_{I}$$

$$\Rightarrow \quad I = \frac{1}{m} - \frac{w^2}{m^2} I$$

$$\Rightarrow \quad I\left(1 + \frac{w^2}{m^2}\right) = \frac{1}{m} \quad \Rightarrow \quad I \times \frac{m^2+w^2}{m^2} = \frac{1}{m} \quad \Rightarrow \quad I = \frac{1}{\cancel{m}} \times \frac{m^2}{m^2+w^2}$$

$$\boxed{I = \frac{m}{m^2+w^2}}$$

Step-2

$$f(x) = \mathcal{F}_c^{-1}\left\{\frac{m}{m^2+w^2}\right\}$$

$$= \frac{2}{\pi}\int_0^\infty \frac{m}{m^2+w^2}\cos(wx)\,dw$$

let, $x = \rho$,

$$f(\rho) = \frac{2}{\pi}\int_0^\infty \frac{m\cos(\rho w)}{m^2+w^2}\,dw$$

$$\frac{\pi}{2} e^{-m\rho} = \int_0^\infty \frac{m \cos(\rho w)}{m^2 + w^2}\, dw$$

let, $m = \beta$,

$$\frac{\pi}{2} e^{-\beta\rho} = \int_0^\infty \frac{\beta \cos(\rho w)}{\beta^2 + w^2}\, dw$$

let, $w = v$,

$$\boxed{\frac{\pi}{2} e^{-\beta\rho} = \int_0^\infty \frac{\beta \cos(\rho v)}{\beta^2 + v^2}\, dv}$$

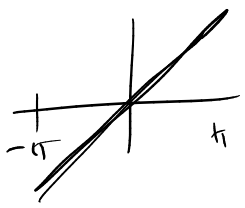
Type-1 (Mistake)

Interval

$$a_k = \frac{1}{L}\int_{-L}^{L} f(x) \cos\left(k \frac{x\pi}{L}\right) dx$$

$f(x) = x$ $\quad \underbrace{(-\pi, \pi)}$

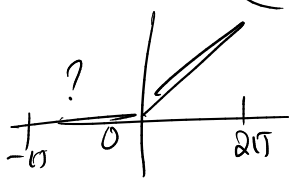
$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \cos(kx)\, dx$$



$2\pi = 2L$

$L = \pi$

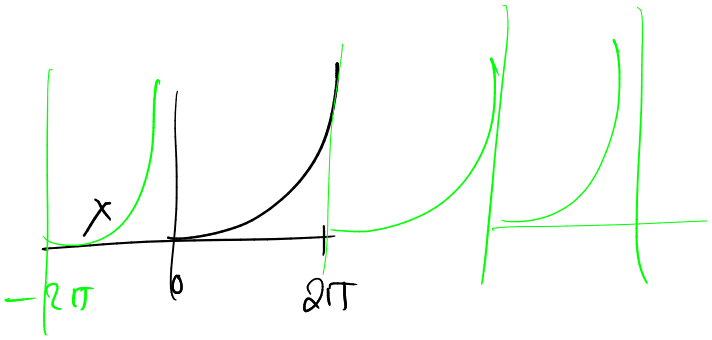
$$a_k = \frac{1}{\pi}\int_0^{2\pi} f(x) \cos(kx)\, dx \longleftarrow (0, 2\pi)$$



Type-2

Even/odd

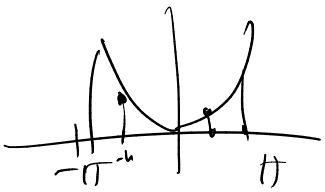
$f(x) = x^2, \quad (0, 2\pi)$



$f(-x) = -f(x)$ or $f(x)$

Even

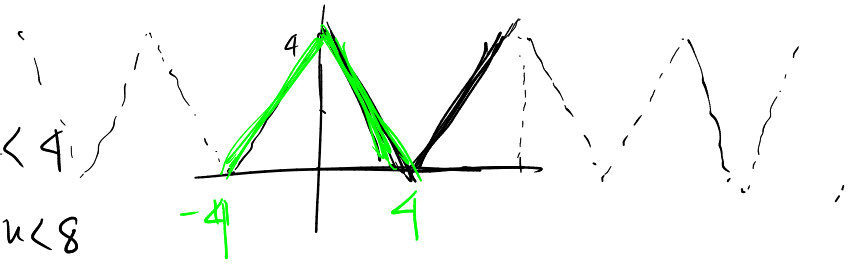
$f(x) = x^2, \quad (-\pi, \pi)$



$f(-x) = f(x)$

Type-3

$$f(x) = \begin{cases} 4-x, & 0 < x < 4 \\ x-4, & 4 < x < 8 \end{cases}$$



$$a_k = \frac{1}{4}\left[\int_0^4 (4-x)\cos\left(k\frac{x\pi}{4}\right)dx + \int_4^8 (x-4)\cos\left(k\frac{x\pi}{4}\right)dx\right]$$

Alternative

$$a_k = \frac{1}{4}\int_{-4}^{4} f(x)\cos\left(k\frac{x\pi}{4}\right)dx$$

$$= \frac{2}{4}\int_0^4 (4-x)\cos\left(k\frac{x\pi}{4}\right)dx$$